Question ID: e86f0651

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

A circle has a radius of 43 meters. What is the area, in square meters, of the circle?

Answer

A. $\frac{43\pi}{2}$

B. 43π

C. 86π

D. $1{,}849\pi$

Correct Answer: D

Rationale

Choice D is correct. The area, A, of a circle is given by the formula $A = \pi r^2$, where r is the radius of the circle. It's given that the circle has a radius of 43 meters. Substituting 43 for r in the formula $A = \pi r^2$ yields $A = \pi(43)^2$, or $A = 1{,}849\pi$. Therefore, the area, in square meters, of the circle is $1{,}849\pi$.

Choice A is incorrect. This is the area, in square meters, of a circle with a radius of $\sqrt{\frac{43}{2}}$ meters.

Choice B is incorrect. This is the area, in square meters, of a circle with a radius of $\sqrt{43}$ meters.

Choice C is incorrect. This is the circumference, in meters, of the circle.